Situation	Recommended plot	Why it helps
Explaining a clinical risk combination to a physician	Highlighted PMF (Case 1 style)	Makes the “exactly k” probability concrete
Justifying sample size for rare adverse events	Complement curve or rare-event PMF	Shows residual risk of missing t event
Inventory / “last unit” decisions	Growth curve of P(at least one)	Reveals diminishing returns
Teaching or onboarding new analysts	Side-by-side comparison + Monte-Carlo	Builds intuition faster than formu alone
Sensitivity analysis	Family of curves for different p p p	Answers “what if the rate is higher/lower?”

Visualizing Binomial Distribution Convergence

The binomial distribution is discrete, yet under two classic limiting regimes it converges to well-known continuous or discrete distributions. These limits are among the most useful approximations in statistics and data science because they let us replace cumbersome binomial calculations with simpler normal or Poisson formulas (or tables) when n is large.

We examine the two most important convergences:

1. **Binomial → Normal** (De Moivre–Laplace / Central Limit Theorem regime)
   When n grows while p stays fixed away from 0 and 1, the shape of the binomial PMF approaches a normal density with the same mean and variance:

$$X \sim \mathrm{Binomial}(n, p) \quad \Rightarrow \quad \frac{X - np}{\sqrt{np(1-p)}} \xrightarrow{d} \mathcal{N}(0, 1)$$

   Rule of thumb: the approximation is usually adequate once $np \geq 5$ and $n(1-p) \geq 5$.

2. **Binomial → Poisson** (rare-event / law of rare events regime)
   When $n \to \infty$ and $p \to 0$ while the mean $\lambda = np$ stays constant,

$$P(X = k) \ \to \ e^{-\lambda} \frac{\lambda^k}{k!}.$$

Both limits appear constantly in practice: normal approximations power many hypothesis tests and confidence intervals; the Poisson limit models rare defects, accidents, mutations, etc.

Practical takeaway

When you need $P(X \leq k)$ for a large binomial (e.g., "probability that at least 55 out of 100 patients respond"), you can safely use the normal CDF (with or without continuity correction) instead of summing hundreds of binomial terms.

The error falls roughly like $1/\sqrt{n}$, exactly as predicted by the local central-limit theorem.

Situation	Limiting regime	Why the approximation helps
200 coin flips – chance of 110–120 heads	Normal	Hand calculation of (200k) \binc {k} (k200) is impossible; normal instantaneous
Number of manufacturing defects in a large batch when defect rate is 0.2 %	Poisson	Software, textbooks and engine speak “Poisson(λ)” language
Daily website visits that convert (conversion rate 1–2 %)	Poisson or Normal	Both limits appear; choose acco whether the mean is small or mo
Number of insurance claims per month	Poisson	Classic actuarial model
Quality-control sampling of 1 000 items	Normal (or Poisson if rare)	Lets you set control limits with μ \mu \pm 3\sigma μ±3σ rules

Quick Reference – When to Use Which Approximation

- Prefer **Normal** when $np \geq 5$ **and** $n(1-p) \geq 5$ (some authors use 10).
- Prefer **Poisson** when $n \geq 20$ (or larger) **and** $p \leq 0.05$ (or $\lambda = np$ is of moderate size).
- When both conditions are borderline, compute the exact binomial (modern computers make this cheap) or use simulation.

The four Python blocks above give you production-ready visual diagnostics. Change `p`, `ns` or `lambda_` and you instantly see how the convergence behaves for any new problem you encounter.